

TC_055: Double_Sending

- **ID:** SCRUM-6
- **Severity:** High
- **Status:** New

Description:

Steps to Reproduce:

1. Log in to the same MoMo account on. Device A and Device B simultaneously
2. Ensure wallet balance = exactly 100,000đ (only enough for 1 transaction)
3. On Device A: start a transfer of 100,000đ
4. Almost simultaneously, on Device B: start a transfer of 100,000đ
5. Confirm both transactions at nearly the same time
6. Observe the result on both devices and check final wallet balance

Test Data:

- Balance: 100,000đ
- Device A: transfer 100,000đ
- Device B: transfer 100,000đ (same time)

Expected Result:

Only 1 of the 2 transactions should succeed. The other should be rejected due to insufficient balance. Wallet balance must never go negative or be deducted twice.

Actual Result:

(conceptual test case — to fill in once tested on live app)

Impact:

Critical security/financial vulnerability — if this occurs, an attacker could exploit the race condition to withdraw more than the actual available balance, causing direct financial loss.

Suggested Fix:

Server-side should use database transaction lock / atomic balance check-and-deduct (e.g. optimistic locking or SELECT FOR UPDATE) instead of checking balance and deducting separately.

Evidence:

Add epic / SCRUM-6

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Simultaneous transfers from 2 devices with the same account may cause balance to go negative (double-spending)

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In Progress

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➦ Improve Bug

▼ Details

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Assignee

👤 Unassigned

Assign to me

Priority

🔴 High

Parent

None

Due date

None

Labels

double-spending

payment,

race-condition,

security,

Team

None

Start date

None

Sprint

SCRUM Sprint 0

Story point estimate

None

Reporter

NT Nguyễn Ngọc Thanh Trúc

Add epic / SCRUM-6

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